UNIVERSITÀ DI BOLOGNA



School of Engineering

Master Degree in Automation Engineering

Computer Vision and Image Processing

Stereo Robot Navigation

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Academic year 2025/2026

Results Analysis

Here we present the results obtained in the stereo-robot navigation project. To carry out this project, we implemented several algorithms and improvements that provide valid solutions, although they differ in result quality. To clearly illustrate these differences, we present plots of the most relevant measurements.

For each method, the following plots are shown:

- **Main Disparities:** Estimates of the mean, mode, and median disparities over time for the frontal area of the environment (with respect to the camera), based on the disparity map of the central region of each reference frame.
- **Distances:** Distance (z, in mm) to the obstacle over time, relative to the moving vehicle, computed from the median disparity (in pixels) obtained from each frame pair.
- **Chessboard Measurements:** Estimated dimensions (W, H in mm) of the chessboard pattern in the scene, along with the corresponding accuracies compared to the known real dimensions.

Chapter 1

ROI Size Selection

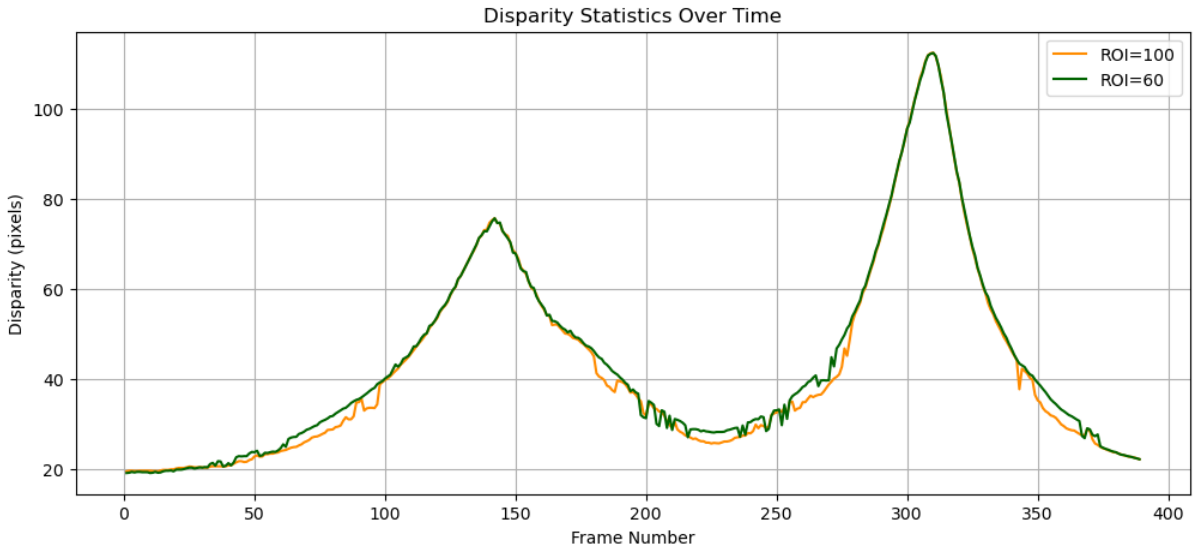


Figure 1.1: Median disparities comparison with size of 60 and 100

Method	**Computational Time (s)**	**Valid Matches**	**Average Width Accuracy (%)**	**Average Height Accuracy (%)**
Naive ROI 60	12.28	612889	92.74	78.09
Naive ROI 100	12.34	2039497	88.96	80.85

Table 1.1: Performance metrics comparison

First, we decided to compare the solutions obtained with different ROI sizes. This parameter determines the portion of the space in front of the stereo camera that is analyzed. In our specific case, the chessboard pattern is located in the central area of each frame; therefore, a smaller ROI size tends to yield better results. However, in a more general stereo navigation task, smaller ROIs may capture fewer image details, potentially reducing accuracy. Hence, selecting an appropriate ROI size represents a trade-off between computational efficiency and depth estimation accuracy.

As expected, a smaller ROI requires less computational time to produce a solution, since there are fewer pixels to match and consequently fewer disparity calculations to perform.

Chapter 2

Gaussian Filter Preprocessing

2.1 Naive solution - No filters

This represents the simplest solution, obtained with an ROI size of 100 and no image filtering applied. It will be used as the benchmark for subsequent comparisons.

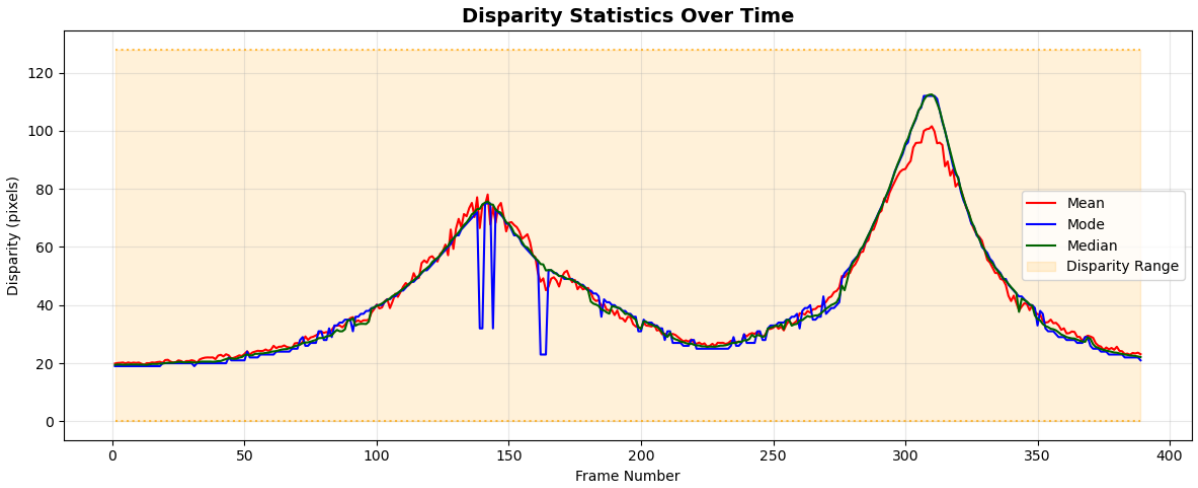


Figure 2.1: disparities for naive solution

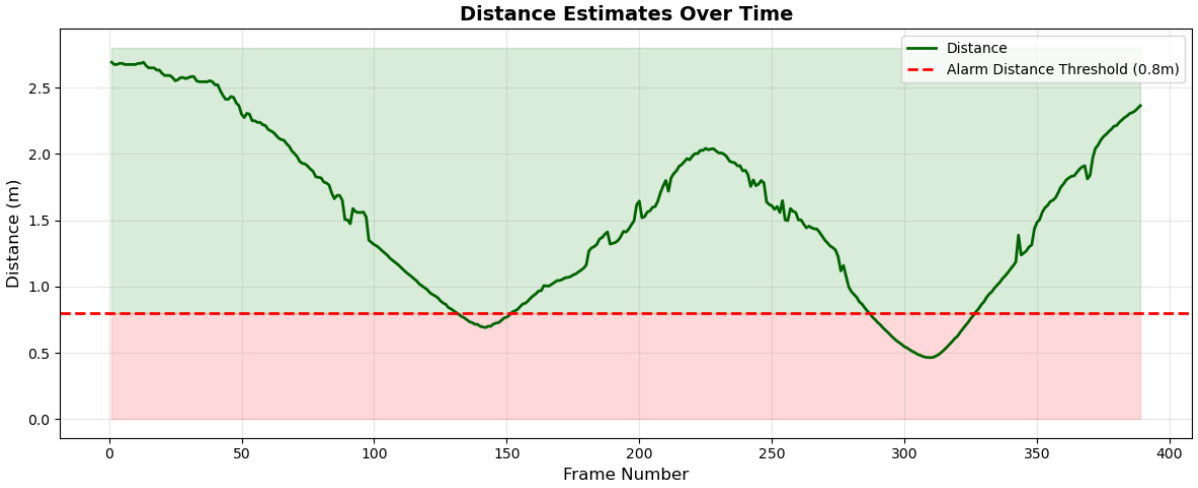


Figure 2.2: distance (z in mm) for naive solution

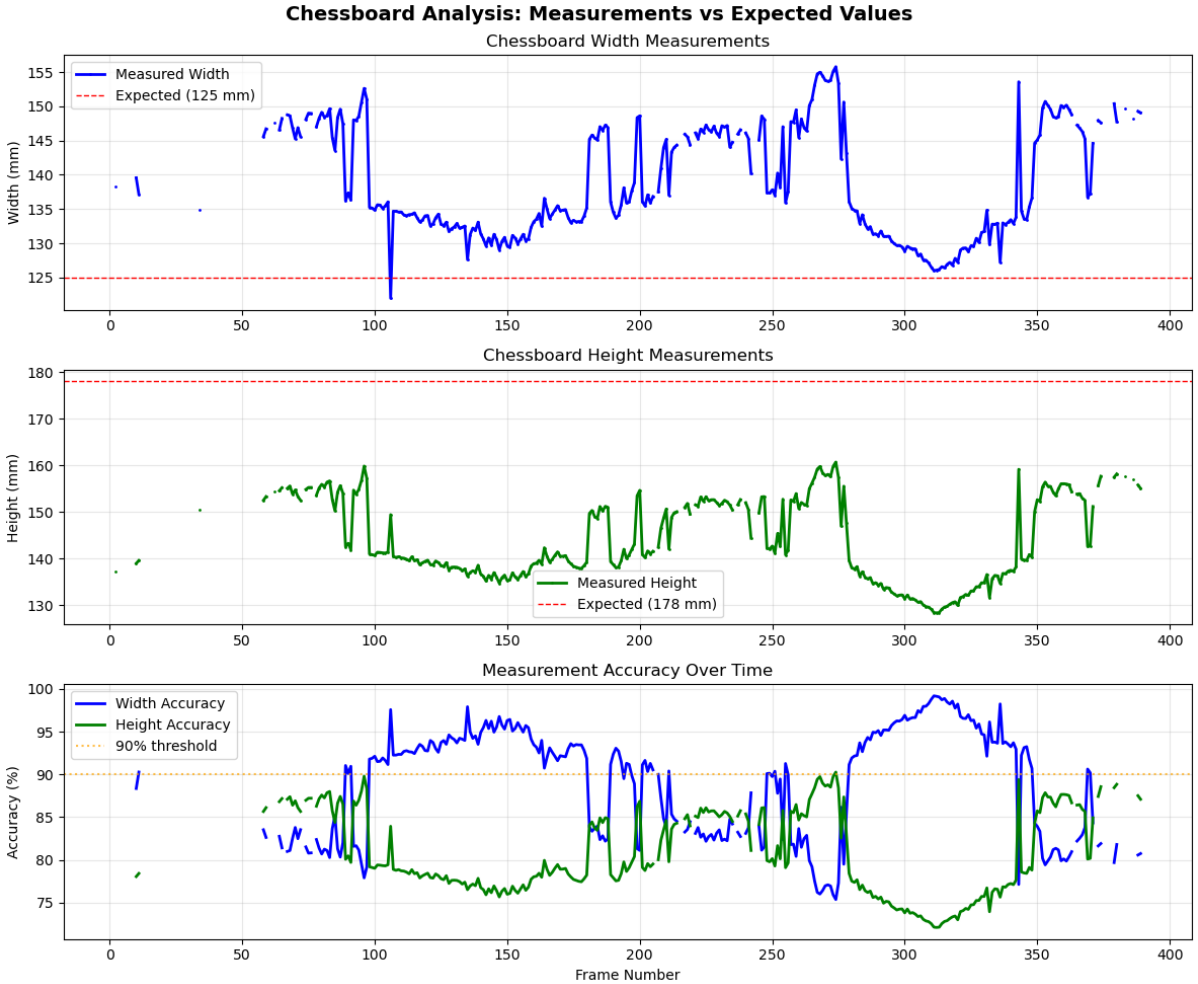


Figure 2.3: chessboard measurements for naive solution

2.2 Gaussian filtered version

To improve the baseline solution, we applied a Gaussian filter to smooth each frame before processing. This filtering step reduces high-frequency noise and enhances the robustness of subsequent feature detection and matching.

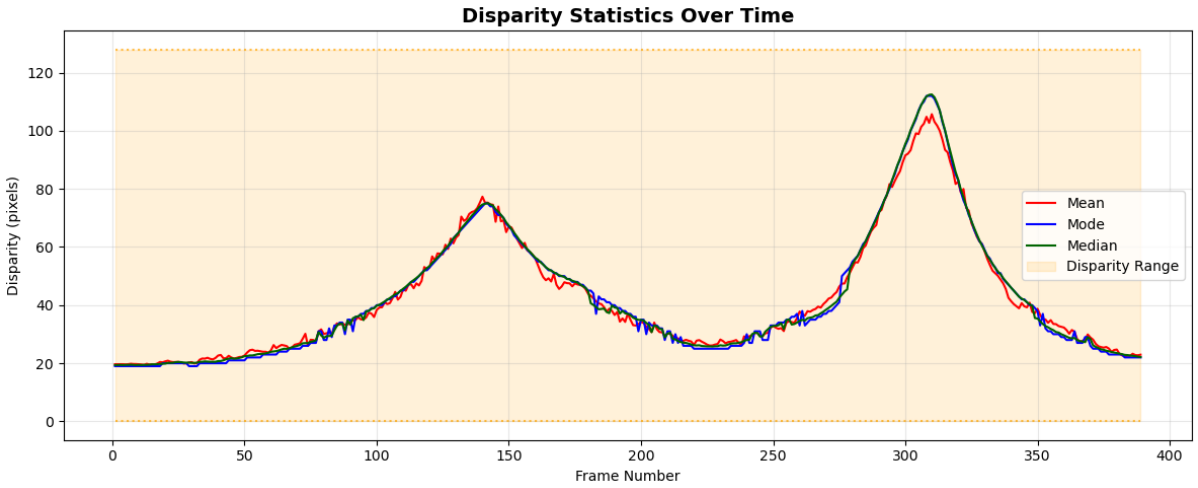


Figure 2.4: disparities with gaussian filter enabled

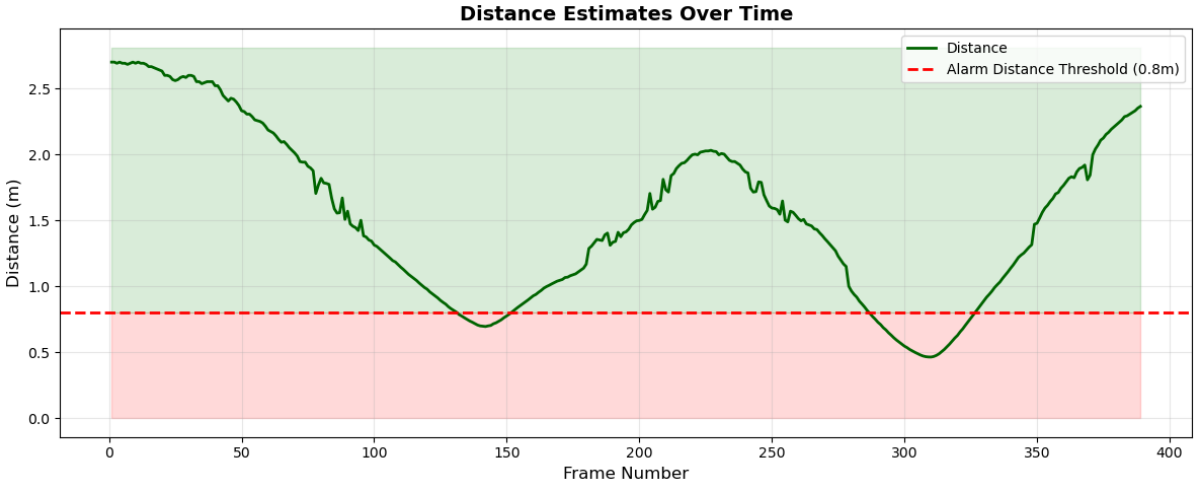


Figure 2.5: distance (z in mm) with gaussian filter enabled

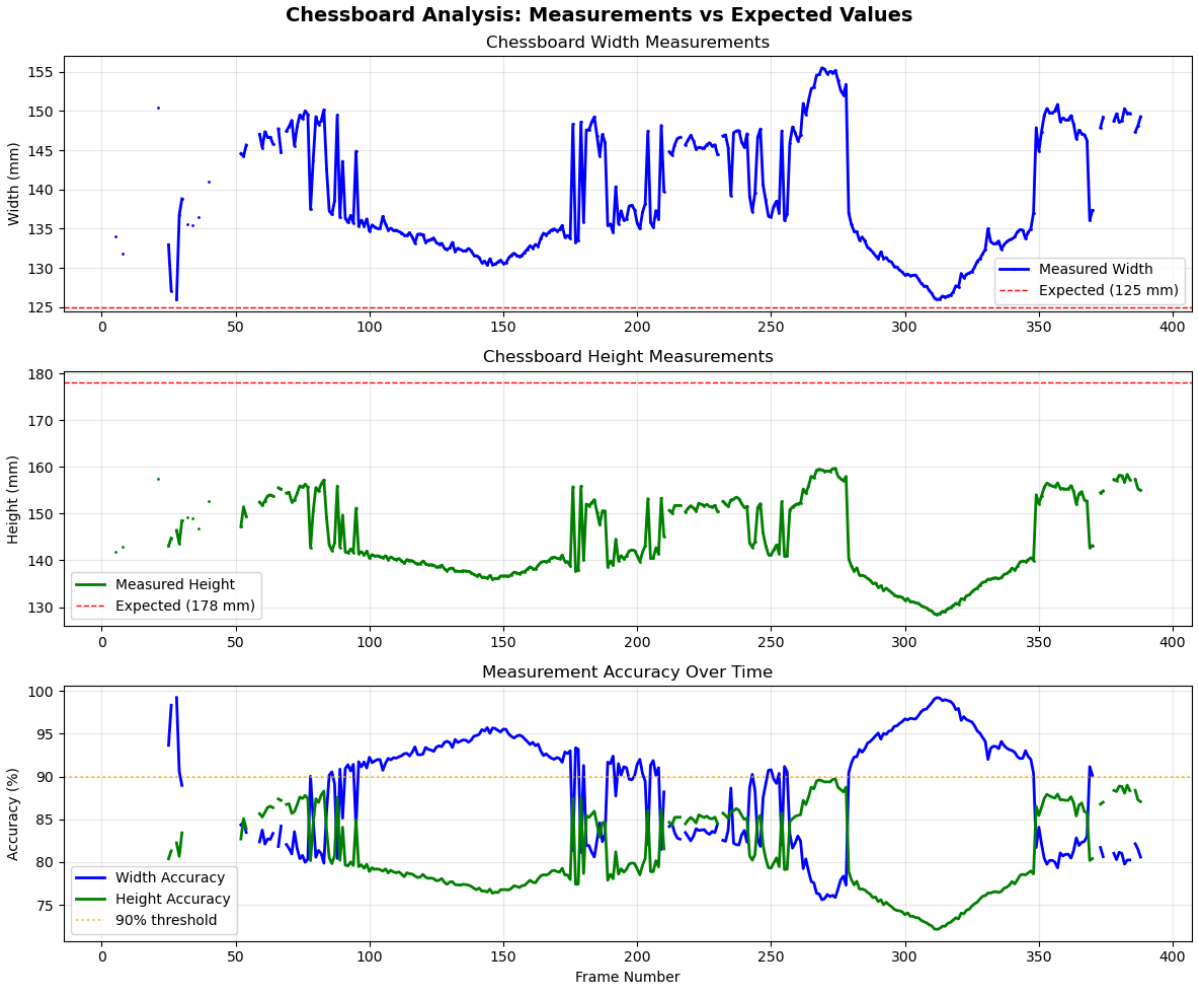


Figure 2.6: Chessboard measurements with gaussian filter enabled

2.3 Naive vs Gaussian filtered methods

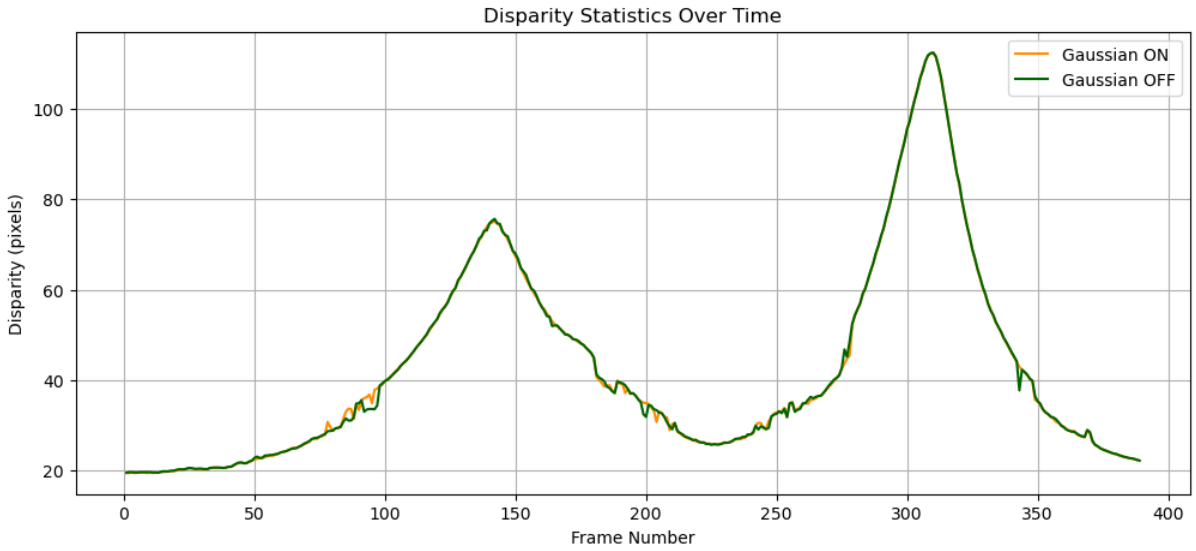


Figure 2.7: Disparity: with and without gaussian filter

Method	**Computational Time (s)**	**Valid Matches**	**Average Width Accuracy (%)**	**Average Height Accuracy (%)**
Gaussian Filter Off	12.34	2039497	88.96	80.85
Gaussian Filter On	10.72	2593965	88.84	81.05

Table 2.1: Performance metrics comparison

As shown in the comparison table 2.1, the total number of valid matches is higher for the solution with the Gaussian filter applied. This improvement occurs because Gaussian blurring removes high-frequency noise that can disrupt block consistency during matching. As a result, the Sum of Absolute Differences (SAD) cost function exhibits clearer minima, leading to more reliable disparity values instead of invalid, zero, or negative outputs. The average accuracies in both width and height, however, remain nearly identical for the two solutions.

Chapter 3

Moravec Interest Operator

3.1 Moravec filtered solution

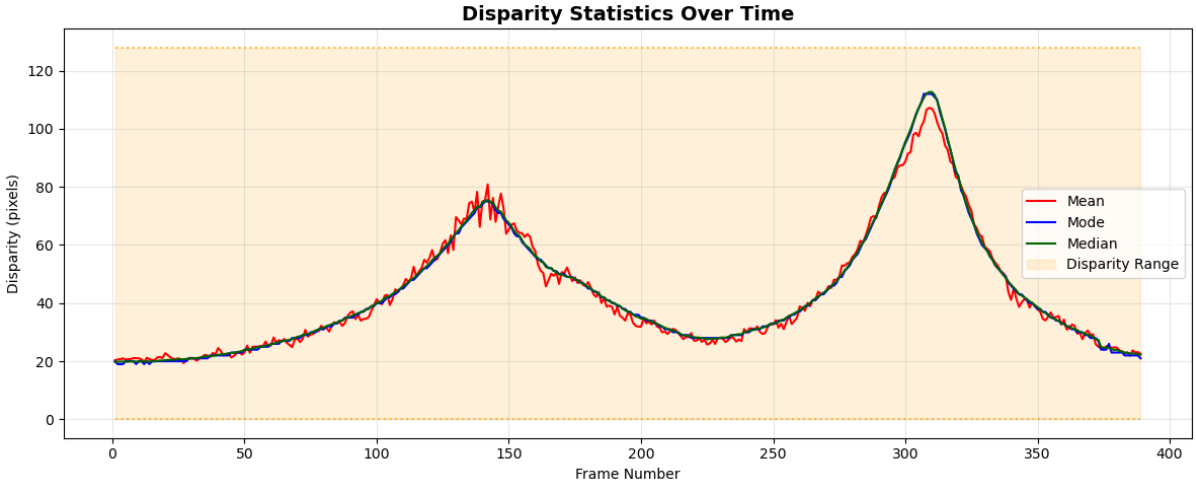


Figure 3.1: Disparities with Moravec filter on, Gaussian filter off

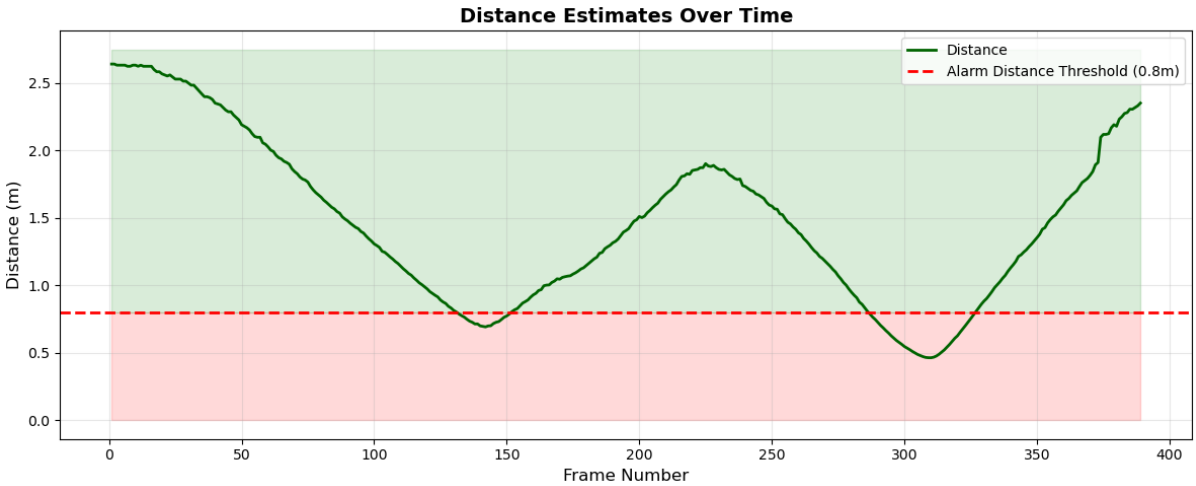


Figure 3.2: Distance (z in mm) with Moravec filter on, Gaussian filter off

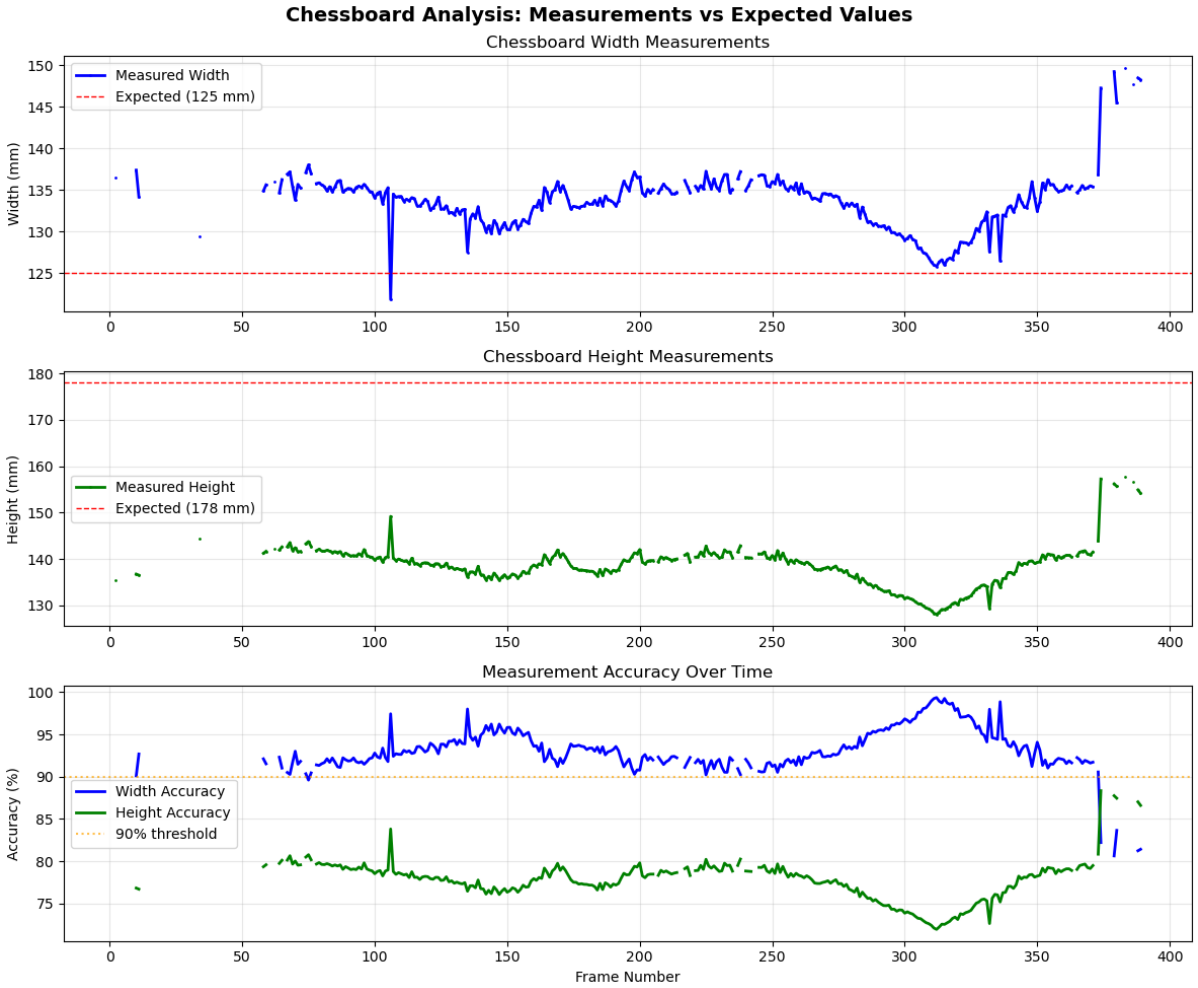


Figure 3.3: Chessboard measurements with Moravec filter on, Gaussian filter off

3.2 Comparison between Moravec and Gaussian filters

3.2.1 Moravec vs Gaussian filtered solutions

The Moravec interest operator filters out outliers in low-texture regions of each frame, improving disparity estimation and increasing the accuracy of the chessboard measurements, though at the cost of higher computational time. As shown in Figure 3.4, the disparity curve over time appears smoother compared to the one obtained using the Gaussian-filtered solution.

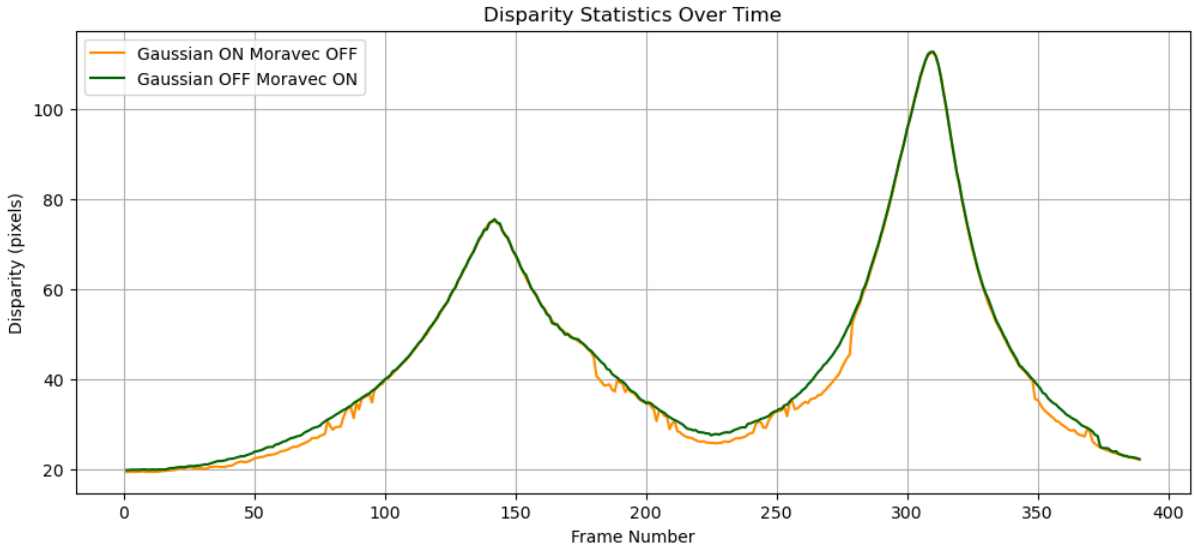


Figure 3.4: Disparity: Moravec filter vs Gaussian filter

3.2.2 Moravec + Gaussian filtered solution

When both the Moravec operator and Gaussian filter are applied during the preprocessing stage, the quality of the results deteriorates. As shown in Figure 3.5, the disparity curve over time exhibits stronger oscillations and spurious spikes. This behavior occurs because Gaussian blurring softens edges and corners, which conflicts with the Moravec operatorâs reliance on local intensity variations to detect textured regions. Since smoothing reduces these critical features, it is advisable not to use both filters simultaneously.

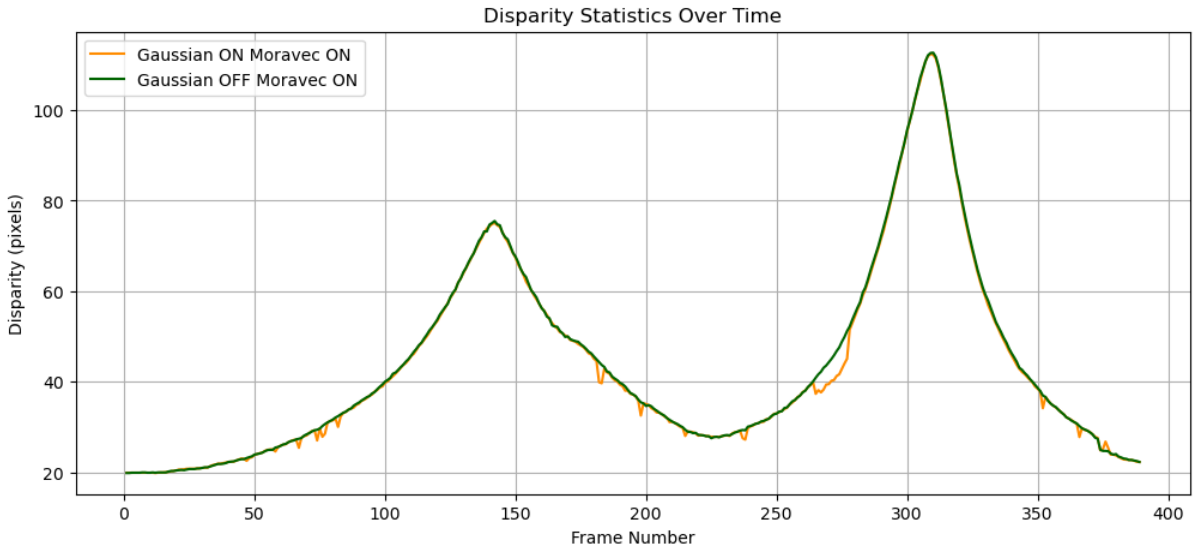


Figure 3.5: Disparity: Moravec filter with and without Gaussian filter

Method	**Computational Time (s)**	**Valid Matches**	**Average Width Accuracy (%)**	**Average Height Accuracy (%)**
Gaussian On, Moravec Off	10.72	2593965	88.84	81.05
Gaussian On, Moravec On	15.85	951935	91.60	79.02
Gaussian Off, Moravec On	17.64	875431	93.08	77.85

Table 3.1: Performance metrics comparison

Chapter 4

Adaptive Disparity Range Improvement

In this chapter, we introduce one of the optional improvements: the use of a dynamic disparity range instead of a fixed one [0, 128]. Specifically, we maintain a disparity range of 64, which is dynamically shifted by an offset. This offset is computed so that the median disparity from the previous frame lies at the center of the current disparity range. Consequently, as the vehicle approaches an obstacle, the horizontal offset increases, preventing the main disparity values from exceeding the valid rangeâand decreases accordingly as the vehicle moves away.

4.1 Dynamic disparity range solution

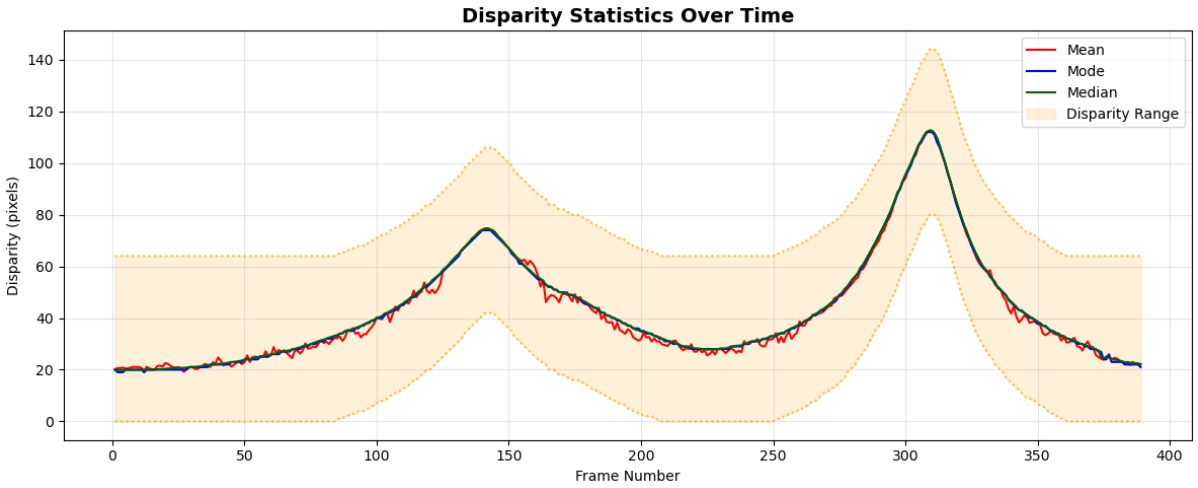


Figure 4.1: Disparities using dynamic disparity range

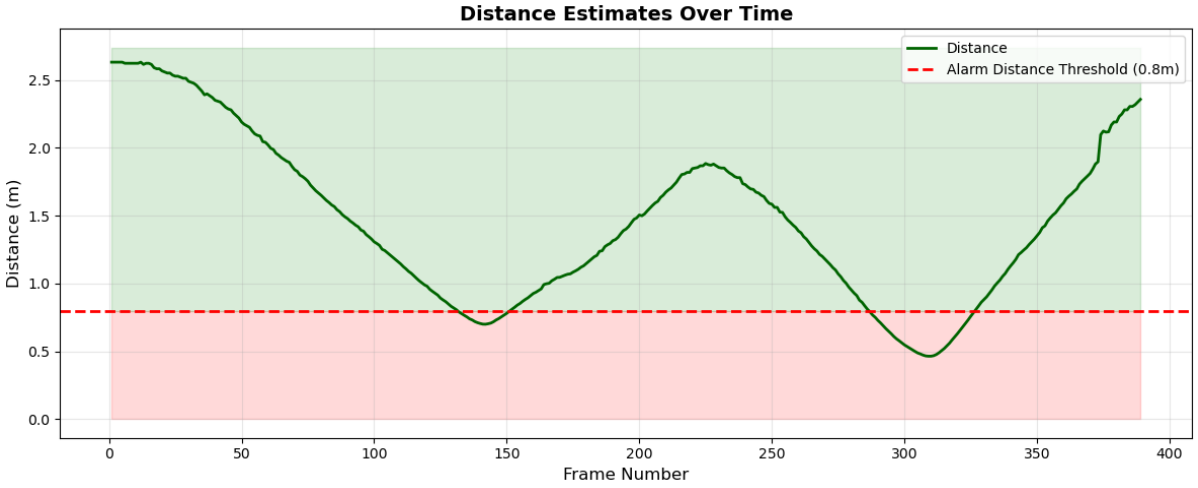


Figure 4.2: Distance (z in mm) using dynamic disparity range

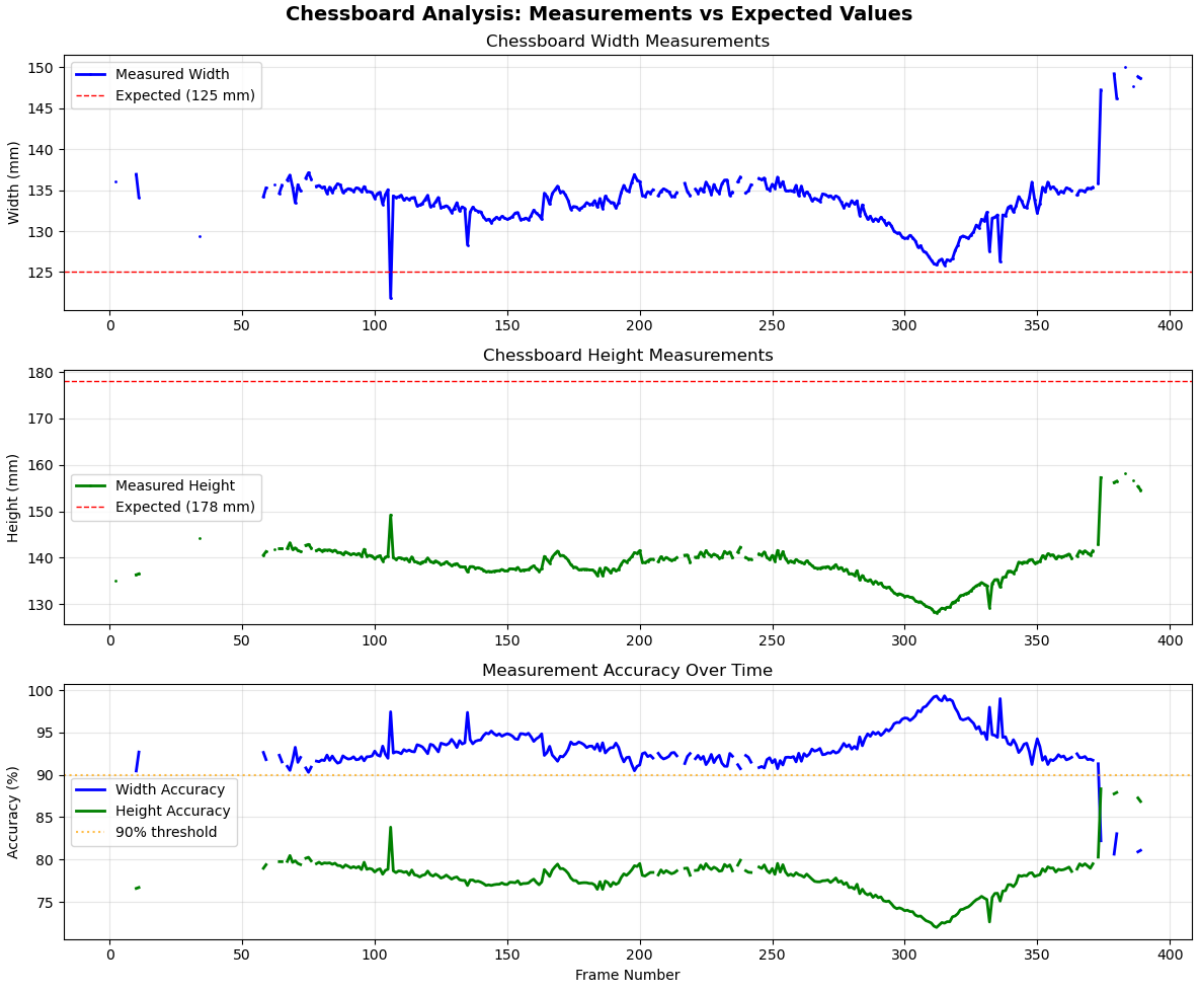


Figure 4.3: Chessboard measurements using dynamic disparity range

4.2 Adaptive vs fixed disparity range

As shown in Table 4.1, the main advantage of this adaptive technique is the reduction in computational complexity, while maintaining the other performance metrics nearly unchanged. This improvement increases system flexibility, ensuring consistent performance both at close range (high disparity values) and at longer distances (low disparity values), without increasing computational effort.

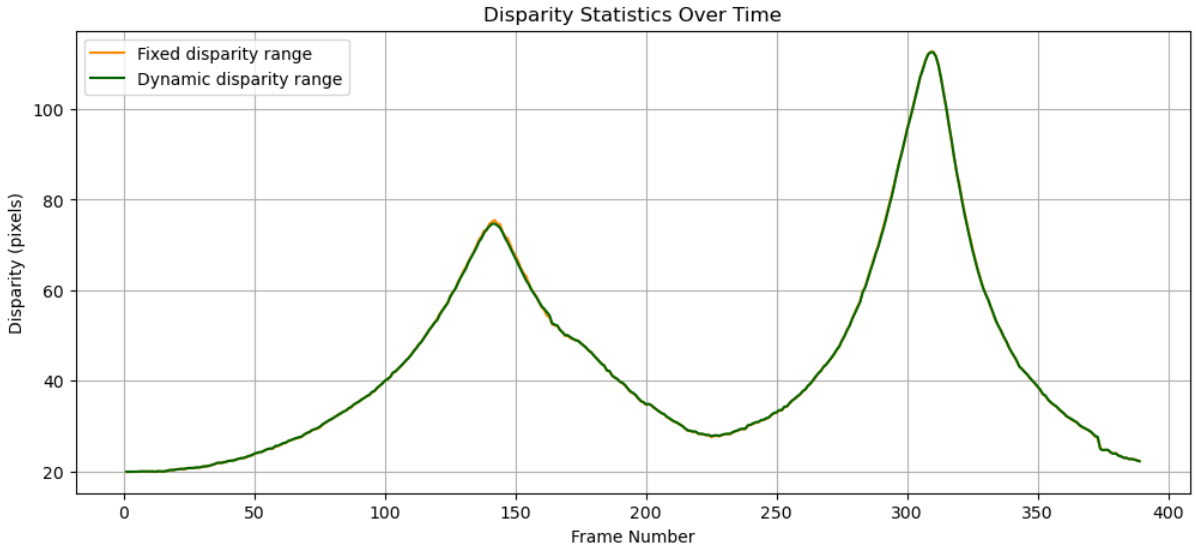


Figure 4.4: Disparity: adaptive vs fixed dynamic range

Method	**Computational Time (s)**	**Valid Matches**	**Average Width Accuracy (%)**	**Average Height Accuracy (%)**
Fixed disparity range [0,128]	17.64	875431	93.08	77.85
Adaptive disparity range [0+o, 64+o]	17.04	873867	93.08	77.84

Table 4.1: Performance metrics comparison dynamic range

Chapter 5

Stripe-Based Planar Analysis

Instead of computing a single main disparity, we estimate a set of disparities corresponding to different regions of the obstacle, allowing us to determine the distance from the camera for each part. The image is divided into five large vertical stripes, and a median disparity value is calculated for each stripe. Based on these values, a planar representation of the obstacle is generated by estimating the angle between the interpolated line connecting the stripe centers and the stereo cameraâs image plane. This approach is particularly useful for estimating the obstacleâs orientation.

5.1 Planar angle estimation

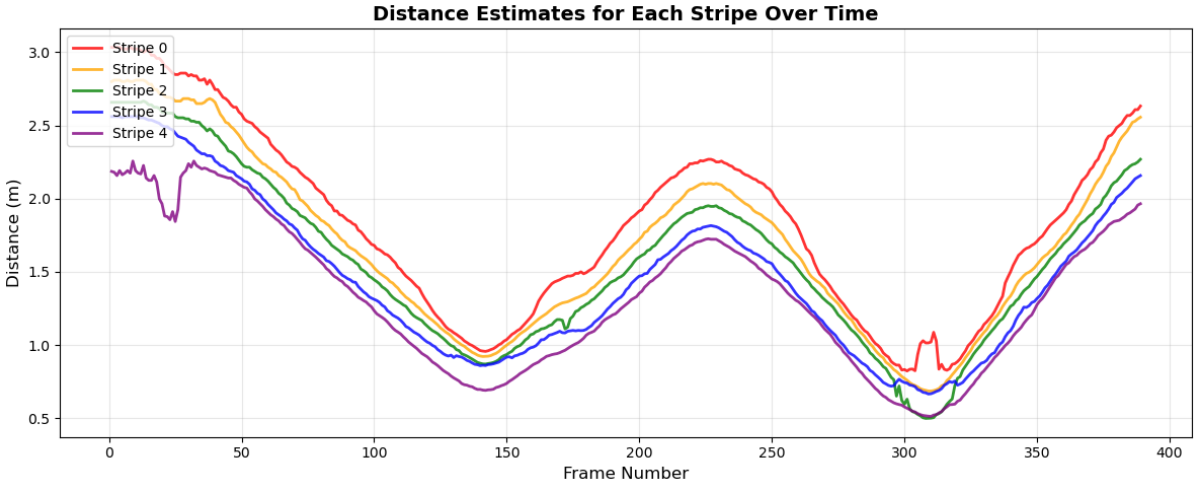


Figure 5.1: distances for each stripe

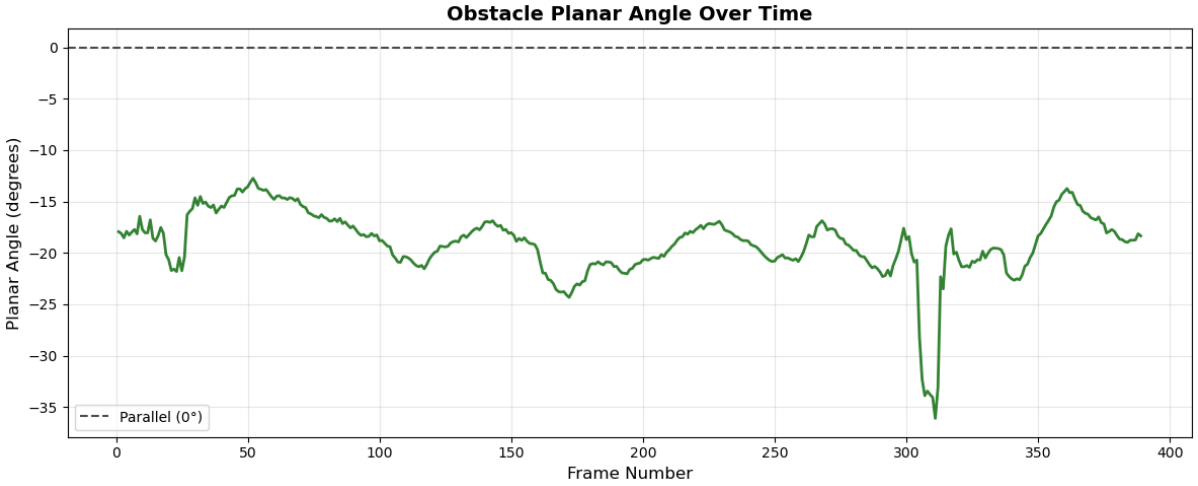


Figure 5.2: Planar angle

5.2 Planar angle estimation with Moravec operator

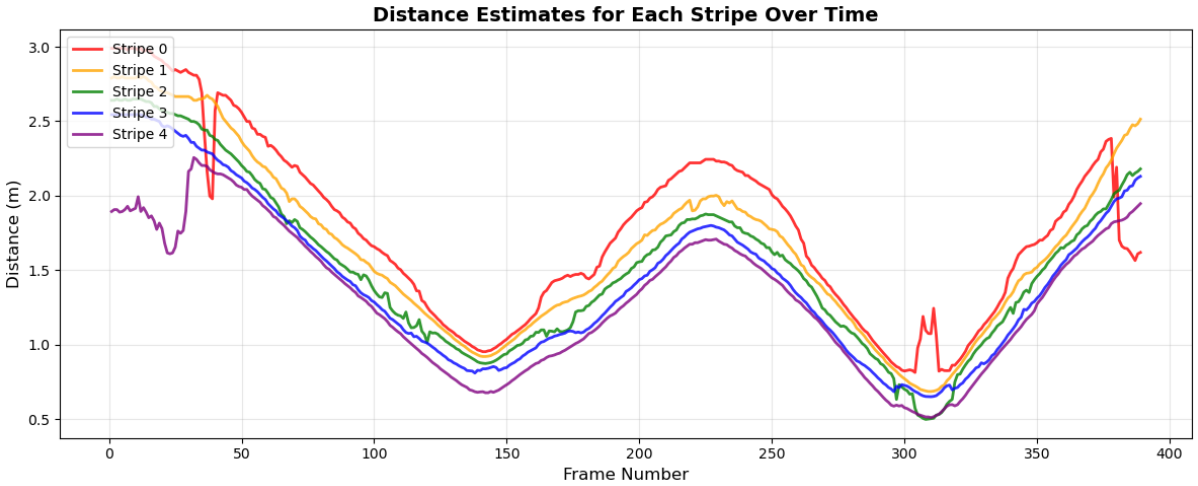


Figure 5.3: distances for each stripe with Moravec filter

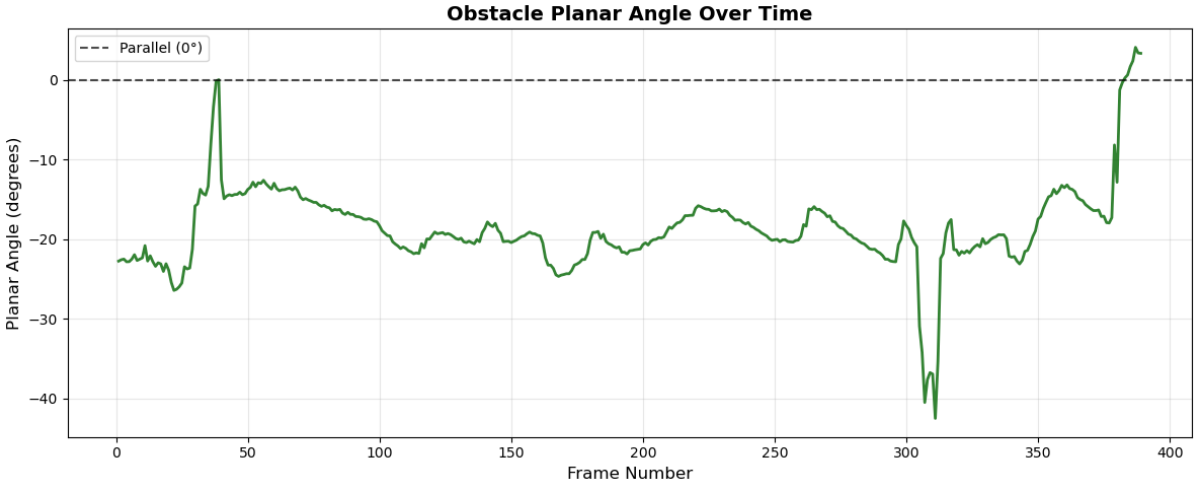


Figure 5.4: Planar angle with Moravec filter

5.3 Planar angle with vs without Moravec filter

As shown in Table **??**, applying the Moravec filter to the full-size image, instead of a smaller ROI, significantly increases computational time.

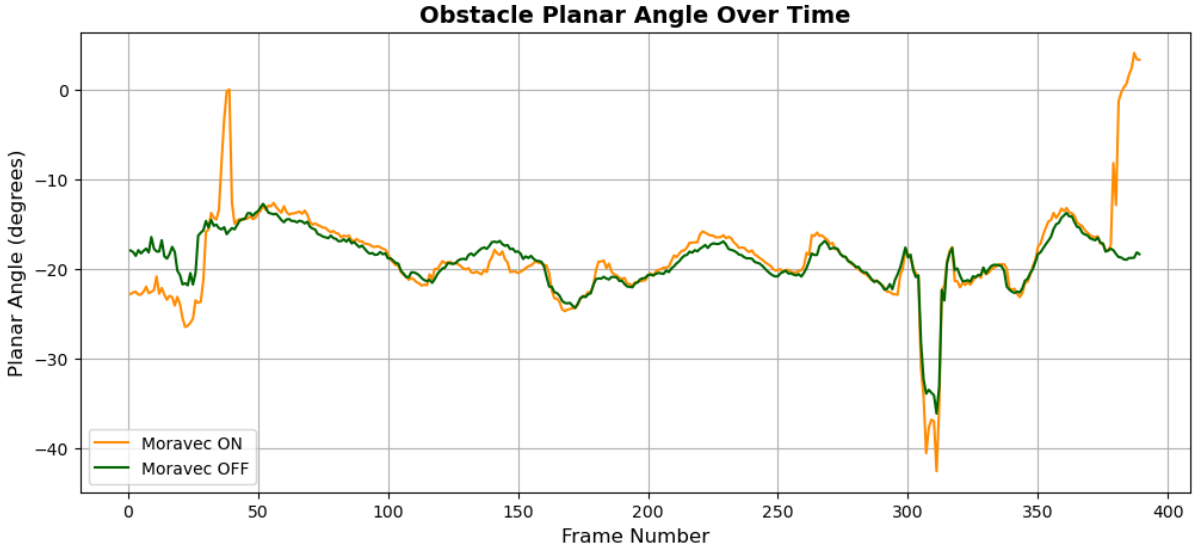


Figure 5.5: Planar angle: Moravec filter on vs off

Method	Computational Time (s)	Average Planar Angle (deg)
Moravec Off	28.51	-19.02
Moravec On	123.44	-18.64

Table 5.1: Performance metrics comparison

Chapter 6

Results Summary

We provide a final summary of the results obtained. As shown in Table 6.1, each implemented solution produces different outcomes. The most appropriate method should be selected based on the specific performance metric of interest

Method	**Computational Time (s)**	**Valid Matches**	**Average Width Accuracy (%)**	**Average Height Accuracy (%)**
Naive ROI 60	12.28	612889	92.74	78.09
Naive ROI 100	12.34	2039497	88.96	80.85
Gaussian filter	10.72	2593965	88.84	81.05
Moravec filter	17.64	875431	93.08	77.85
Adaptive disparity range	17.04	873867	93.08	77.84

Table 6.1: Performance metrics comparison